

BALANCE–FIELD FRAMEWORK (BFG)

The BFG Endogenous Dual-Order Strong Universal Reclosure Equation — Canonical Closure Edition

*A Canonically Self-Closed Mathematical Architecture with Reciprocal Balance,
Spectral Persistence, Polar Capacity Transport, and Stage-Free Natural-System
Realization*

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Scientific

status

Independent theoretical whitepaper and falsifiable research program. Exact neutral identities, reciprocal-order uniqueness, polar-support transport, positivity reconstruction, and the dual-order non-expansion inequality are mathematical results under the declared admissibility assumptions. The Canonical Endogenous Dual-Order Strong Universal Reclosure construction is mathematically universal over its totalized BFG operator category: one fixed generator produces target-independent invariant classes, an internally derived one-degree-of-freedom order state, canonical persistent selection, canonical next-support transport, and terminal failure without stage-specific repair. The internal universal map contains no natural-stage predicate, externally supplied order coordinate, tunable persistence contour, or free synthesis matrix. Natural-system identification with physical, chemical, biological, neural, conscious, or phenomenal regimes remains a separately falsifiable realization hypothesis. Structural closure loads Λ are not asserted to be joule-valued energy. Numerical studies are deterministic or seeded synthetic checks of the formal identities and are not empirical validation.

Abstract

This whitepaper develops the Canonical Closure Edition of the Endogenous Dual-Order Strong Universal Form of the Balance-Field Framework (BFG). The aim is to remove avoidable arbitrariness from the universal generator rather than to enlarge it with additional sector-specific structure. Starting from Level-0 admissible formation, the construction uses positive master geometry, the exact neutral resolvent pair $C_N(Y)=(I+Y)^{-1}$ and $B_N(Y)=Y(I+Y)^{-1}$, recursive persistence, and totalized reclosure. The dual-order state is not introduced as an independent fitting device. The exact neutral pair produces two complementary persistent branches, keep and upward. Their nonnegative closure loads Λ^{keep} and $\Lambda^{\uparrow}$ determine the unique normalized reciprocal weights ω^{keep} and $\omega^{\uparrow}$ satisfying the balance equation $\omega^{\text{keep}}\Lambda^{\text{keep}}=\omega^{\uparrow}\Lambda^{\uparrow}$ together with $\omega^{\text{keep}}+\omega^{\uparrow}=1$. Hence the dual-order pair is a one-degree-of-freedom projective representation of the closure-load ratio, not two new free parameters. Persistence is defined canonically by bounded non-decaying recursive modes; the Riesz projector is only a representation of the resulting isolated peripheral spectral subspace and does not introduce a tunable contour choice. The next-level synthesis map is fixed by the polar

decomposition of the cross-fed analysis operator, eliminating a free capacity-transport map. Positive master geometry is then rebuilt by the same Gram construction and the same formation gate at every nonterminal iterate. The resulting EDO-URE is a totalized self-map $\mathbb{U}$ on the declared admissible BFG operator category. The construction extends from finite-dimensional matrices to separable Hilbert-space objects through the nonnegative self-adjoint form $Y=K^\dagger K$ and its bounded resolvents. Named physical, chemical, biological, cognitive, conscious, and phenomenal regimes remain outside the generator and enter only through independently testable realization maps. Analytical results establish reciprocal-order uniqueness, canonical transport on the active support, positivity preservation, exact neutral partition, and local non-expansion. New deterministic and 10,000-instance complex-matrix tests verify the numerical consequences of those theorems to machine precision. The paper therefore closes the principal mathematical degrees of freedom inside the declared universal category while keeping empirical realization claims independently falsifiable rather than hiding them inside the formalism.

Keywords: Balance-Field Framework; BFG; endogenous dual order; EDO-URE; reciprocal balance; canonical order; strong universality; universal reclosure; Level-0; neutral resolvent; positive master geometry; spectral persistence; polar decomposition; canonical capacity transport; recursive persistence; no-retuning; invariant quotient; natural realization; consciousness; qualia.

Contribution and source-line provenance

The Strong Universal Form, the totalized admissible state category, target-independent invariant quotient, natural-realization commutation test, universal no-retuning criterion, and the EDO-URE are integrated here by Marcel Theodor Wende. This Canonical Closure Edition strengthens the construction by deriving the dual-order pair from an exact reciprocal-balance condition, defining persistence by a target-independent bounded non-decay criterion, replacing any free synthesis map by the polar partial isometry of the cross-fed analysis operator, and extending the resolvent core to the natural quadratic-form Hilbert setting. Earlier co-authored source lines remain credited numerically where their mathematics is used; the reciprocal-balance uniqueness theorem, canonical polar transport, and the resulting closure certificate are new unification results of this paper.

- Marcel Theodor Wende and Thomas W. Loker: co-authored Level-0 foundation source line used for pre-polar admissibility, determinate formation, and the Level-0 boundary [3].
- Marcel Theodor Wende and Luke Kenneth Casson Leighton: recursive conjugate-closure source line, including the form $R_C=MRM^\dagger$ with $M=\sqrt{N}$ used in the persistence architecture [4].

Reference and numbering convention

This whitepaper uses a single numerical reference sequence in order of first substantive appearance, with bracketed in-text citations and one numbered bibliography entry per source. Equations are numbered globally and consecutively. Figures and tables are numbered independently. Internal BFG documents are named only in the numbered bibliography; the body text uses source-line descriptions and numerical citations.

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1. Purpose, scope, and the unification criterion

The target is mathematical universality in a precise sense: one closed operator architecture must regenerate the structural data needed for its own next application without importing a stage label, a stage-specific order law, a tunable persistence selector, or a freely chosen next-level synthesis map. The criterion is therefore stronger than repeated use of the same notation. Universality requires a self-map on one declared admissible category, canonical internal construction of every generator-level coordinate, an explicit terminal state for failed generation, and target-independent invariant classes. Empirical universality is a separate realization claim and is tested only after the mathematical classes have been constructed.

The Canonical Closure Edition focuses on four potential sources of hidden freedom: order, persistence selection, capacity transport, and carrier dimension. Order is fixed by a reciprocal-balance theorem; persistence is the bounded non-decaying recursive subspace rather than a hand-selected spectral window; transport is the polar partial isometry of a canonically determined analysis operator; and the neutral resolvent core is formulated so that nonnegative self-adjoint loads on separable Hilbert spaces remain admissible. No named natural regime participates in any of these definitions.

$$\mathcal{T}_0 \rightarrow \Delta N_0 \rightarrow (P_1, N, P_2) \rightarrow K_0 \rightarrow \delta^* \rightarrow H_C \rightarrow Y_C \rightarrow \{C_N, B_N\} \rightarrow R_C \rightarrow \mathbb{U} \rightarrow \mathbb{U}^2 \rightarrow \quad (1)$$

...

Equation (1) remains the red thread, but the iterated engine is now $\mathbb{U}$ rather than a recursion plus an externally supplied scalar order coordinate. The map is completed below as a totalized, level-independent self-map whose output carries the same admissible structural data needed for the next iteration. Named natural regimes are downstream realization hypotheses, never terms in the generator.

Unification	criterion
An alleged stage transition is not counted as unified if it requires a new sector-specific fundamental response function, a post-hoc role redefinition, a target-aware threshold, or a carrier map chosen only after the higher-level outcome is known.	

2. Epistemic discipline and claim-state hierarchy

The claim-state hierarchy is retained rigorously [1,2]. Definitions specify the state category and equivalence relation; inherited structural axioms supply Level-0, exact neutral resolution, positive closure geometry, and recursive persistence; the Strong Universal completion specifies how these structures are transported under reclosure; the mathematical theorem is conditional on those declared axioms; physical principles such as local physical closure remain separate; and all identifications with natural systems remain empirical hypotheses.

Table 1. Claim-state discipline used throughout the paper.

Class	Meaning
Definition	Introduces an object or equivalence class; true only by stipulation within the formalism.
Axiom / inherited BFG structure	A structural assumption imported from the canonical corpus.
Theorem	Exact mathematical consequence under explicitly stated assumptions.
Conditional theorem	Exact consequence after additional assumptions not yet derived upstream.
Physical principle	A universal physical bridge constrained by no-retuning; LPCP is the canonical example [1].
Unification construction	New in this paper: a fixed self-reclosing operator recursion, including endogenous dual-order feedback, built only from inherited canonical objects and explicitly declared universal constructions.
Model completion	A concrete representational or toy choice used to make a calculation executable.
Spectral prescription	A fixed readout from the master spectrum; numerical content must be computed.
Empirical hypothesis	Identification of a formal closure class with an observed physical or phenomenal regime.
Synthetic simulation	Numerically exact for the stated toy model; not empirical validation.

The strongest formal conclusion is therefore not that every natural object has already been derived. It is that, once the Strong Universal admissible category is fixed, the same target-independent map closes on that category and generates only domain-independent invariant classes. The separate burden is to show that atoms, cells, organisms, conscious states, or other natural systems realize those classes under prospectively fixed measurement maps.

3. Canonical source architecture

3.1 Level-0 trinitarian source

The canonical Level-0 formulation defines a finite-dimensional real Hilbert space V_0 with three primitive admissibility roles: difference capacity $\mathfrak{D}_0$, coherence capacity $\mathfrak{c}_0$, and neutral admissibility $\mathfrak{N}_0$ [1,3].

$$\mathcal{T}_0 = (\mathfrak{D}_0, \mathfrak{c}_0, \mathfrak{N}_0) \tag{2}$$

$$K_0 = c_0 + N_0 - \mathcal{D}_0 \quad (3)$$

$$\mathcal{A}_{\text{Adm}}(\delta) = \frac{1}{2}\langle \delta, K_0 \delta \rangle + \frac{1}{4}\|\delta\|^4, \quad \delta \in V_0 \quad (4)$$

A nontrivial isolated formation branch requires a simple negative minimum eigenvalue of K_0 and a positive spectral gap.

$$\lambda_0 = \min \sigma(K_0) < 0, \quad \text{mult}(\lambda_0)=1 \quad (5)$$

$$\Delta_0 = \lambda_1 - \lambda_0 > 0 \quad (6)$$

$$\delta^*_{\pm} = \pm\sqrt{-\lambda_0}e_0 \quad (7)$$

The quartic term makes the admissibility functional coercive; the Rayleigh–Ritz argument isolates the lowest eigenvector. The two minima are polarity-related. This is a mathematical formation result inside the stated real finite-dimensional model, not an empirical cosmological timestamp [1–3].

3.2 Exact neutral elimination

Let A map a difference channel into a neutral mediation channel. The canonical closure formulation uses a quadratic neutral completion whose auxiliary neutral variable can be eliminated exactly [1].

$$\mathcal{A}_C(D, v) = \frac{1}{2}\|D - A^\dagger v\|^2 + \frac{1}{2}\|v\|^2 \quad (8)$$

$$Y = A^\dagger A \geq 0 \quad (9)$$

$$v^* = (I + AA^\dagger)^{-1}AD \quad (10)$$

$$C_N(Y) = (I + Y)^{-1} \quad (11)$$

$$B_N(Y) = Y(I + Y)^{-1} = I - C_N(Y) \quad (12)$$

$$C_N(Y) + B_N(Y) = I \quad (13)$$

$$0 < C_N(Y) \leq I, \quad 0 \leq B_N(Y) < I \quad (14)$$

$$\mathcal{A}_C^{\wedge \min}(D) = \frac{1}{2}\langle D, C_N(Y)D \rangle \quad (15)$$

$$\frac{1}{2}\|D\|^2 = \frac{1}{2}\langle D, C_N D \rangle + \frac{1}{2}\langle D, B_N D \rangle \quad (16)$$

Equations (9)–(16) are the central exact identities of the unification. They supply a bounded retained response, a bounded complementary response, and an exact quadratic partition. Standard positive-matrix analysis supports the spectral and perturbative facts used here [7,8].

3.3 Positive master closure geometry and spectral sectors

The canonical universal closure architecture lifts the local neutral pair into a positive master closure geometry [1].

$$B_C = D_{\text{cov}} c \quad (17)$$

$$H_C = B_C^\dagger W_N B_C \geq 0 \quad (18)$$

$$Y_C = L_C^\dagger H_C L_C \geq 0 \quad (19)$$

Sector identity is licensed only by upstream spectral gaps. If S_C is the relevant self-adjoint master sector operator and Δ_a are disjoint spectral windows, then:

$$P_a = \mathbb{1}_{\{\Delta_a\}}(S_C), \quad P_a P_b = \delta_{ab} P_a, \quad \Sigma_a P_a = I \quad (20)$$

$$\mathcal{B}_a = B_N^{\wedge \{1/2\}} P_a B_N^{\wedge \{1/2\}}, \quad \mathcal{B}_a \geq 0, \quad \Sigma_a \mathcal{B}_a = B_N \quad (21)$$

This positive partition is the exact no-double-counting backbone of later readouts. A claimed higher-stage channel that has no upstream gap or invariant distinction is not permitted to become a new fundamental BFG sector simply by naming it.

3.4 Recursive persistence and source-line continuity

The recursive conjugate-closure source line is preserved explicitly. Its conjugate form is used in the persistence architecture [4]. The canonical universal architecture uses a protected persistence direction plus a stable complement [1].

$$R_C = M U_Ω C_E M^\dagger, \quad M = \sqrt{N_R} \quad (22)$$

$$R_C e_T = e_T \quad (23)$$

$$\|R_C|_{E_s}\|_L \leq r < 1 \quad (24)$$

$$P_T = \text{RieszProj}_{\{\lambda=1\}}(Dc_phys) \quad (25)$$

For $s = ae_T + s_s$, the stable component contracts at least geometrically, so the full trajectory remains power bounded. This motivates the carrier-neutral witness notion used below. The mathematical distinction between eigenvalue information and robust subspace information follows standard perturbation theory [7,8].

4. Why endogenous dual order is canonical rather than ad hoc

4.1 Structural origin of the dual pair

The dual order state arises only after the exact neutral pair has already produced two mathematically distinct but complementary persistent responses. BFG therefore does not begin by postulating two order parameters. It begins with one retained branch and one complementary upward branch generated by C_N and B_N . The order problem is then: how can one assign a normalized internal mediation state to these two branches without inserting a new scale, a new sector coefficient, or a self-reinforcing preference for the already dominant branch? The answer used here is reciprocal balance: each branch receives the weight required to equalize its order-weighted closure load with the other branch. This is the minimal negative-feedback completion of a two-pole balance geometry.

4.2 Canonical persistent geometry

Let Y_n be the positive master load of the current state. Its strictly positive graph metric G_n defines the norm in which persistent projection and closure loads are measured.

$$G_n := I + Y_n > 0, \quad \|z\|_{G_n}^2 := \langle z, G_n z \rangle \quad (26)$$

Persistence itself is not selected by choosing a convenient contour. The witness space is defined by a target-independent dynamical criterion: recursive trajectories must remain bounded and must not asymptotically disappear. In the finite-dimensional power-bounded case with semisimple peripheral spectrum this is exactly the direct sum of the unit-modulus eigenspaces.

$$\begin{aligned} \mathcal{W}_n &:= \text{cl span}\{z : \sup_{k \geq 0} \|R_{\{C,n\}}^k z\|_{G_n} < \infty, \limsup_{k \rightarrow \infty} \|R_{\{C,n\}}^k z\|_{G_n} > 0\} \\ &= \bigoplus_{\{|\lambda|=1\}} \ker(R_{\{C,n\}} - \lambda I) \quad (\text{finite-} \\ &\quad \text{dimensional power-bounded case}) \end{aligned} \quad (27)$$

When the peripheral block is isolated from the strictly decaying complement by a positive spectral gap, its Riesz projection is unique. Any contour enclosing exactly that isolated set yields the same projector; the contour is therefore a computational representation of a previously defined subspace, not a tunable model parameter [7–10].

$$\begin{aligned} \gamma_n &:= 1 - \sup_{\{\lambda \in \sigma(R_{\{C,n\}}) \setminus \sigma_{\text{per}}\}} |\lambda| > 0, \\ P_n^R &:= (1/2\pi i) \oint_{\Gamma(\sigma_{\text{per}})} (zI - R_{\{C,n\}})^{-1} dz \end{aligned} \quad (28)$$

$$dz, \quad \mathcal{W}_n = \text{Ran}(P_n \wedge R)$$

If W_n is any full-column-rank basis of $\mathcal{W}_n$, the master metric gives the unique G_n -orthogonal projector. The result is basis independent, idempotent, G_n -self-adjoint, and contractive in the G_n norm.

$$\begin{aligned} P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}} &:= \\ W_n(W_n^\dagger G_n W_n)^{-1} W_n^\dagger G_n, \\ (P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}})^2 &= P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}}, \\ (P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}})^\dagger G_n &= G_n P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}} \end{aligned} \tag{29}$$

4.3 Exact keep, upward, and export roles

The persistent projector acts on both exact neutral responses. Keep, upward, and export are therefore roles derived from one partition and one persistence criterion, not independent gates.

$$D_n^{\wedge\{\text{keep}\}} = P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}} C_N(Y_n) D_n, \tag{30}$$

$$D_n^{\wedge\{\uparrow\}} = P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}} B_N(Y_n) D_n,$$

$$D_n^{\wedge\{\text{out}\}} = (I - P_{\{\mathcal{W}_n\}^{\wedge}\{(G)\}}) B_N(Y_n) D_n$$

$$\Lambda_n^{\wedge\{\text{keep}\}} := \|D_n^{\wedge\{\text{keep}\}}\|_{G_n}^2, \tag{31}$$

$$\Lambda_n^{\wedge\{\uparrow\}} := \|D_n^{\wedge\{\uparrow\}}\|_{G_n}^2,$$

$$S_n := \Lambda_n^{\wedge\{\text{keep}\}} + \Lambda_n^{\wedge\{\uparrow\}}$$

4.4 Reciprocal Balance Principle and uniqueness

The dual-order map is fixed by two equations rather than chosen from a convenient functional family. First, the two weights must be a normalized positive partition. Second, the order-weighted closure loads must be equal. The second condition is the operational balance statement: neither persistent pole is permitted to dominate the next closure merely because its present load is larger.

$$\begin{aligned} \omega_n^{\wedge\{\text{keep}\}} \geq 0, \quad \omega_n^{\wedge\{\uparrow\}} \geq 0, \quad \omega_n^{\wedge\{\text{keep}\}} \\ + \omega_n^{\wedge\{\uparrow\}} = 1, \\ \omega_n^{\wedge\{\text{keep}\}} \Lambda_n^{\wedge\{\text{keep}\}} = \omega_n^{\wedge\{\uparrow\}} \Lambda_n^{\wedge\{\uparrow\}} \end{aligned} \tag{32}$$

Proposition — Reciprocal-order uniqueness. For $\Lambda_n^{\wedge\{\text{keep}\}} > 0$ and $\Lambda_n^{\wedge\{\uparrow\}} > 0$, Equation (32) has exactly one solution on the one-simplex. Solving the two linear constraints gives the cross-fed pair below. Thus the reciprocal form is not an extra empirical fit and contains no exponent, gain, threshold, or sector-specific coefficient.

$$\begin{aligned} \omega_{\{U,n+1\}} &:= (\omega_{\{n+1\}}^{\wedge\{\text{keep}\}}, \omega_{\{n+1\}}^{\wedge\{\uparrow\}}) \\ &= (\Lambda_n^{\wedge\{\uparrow\}}/S_n, \Lambda_n^{\wedge\{\text{keep}\}}/S_n) \in \Delta^1 \end{aligned} \tag{33}$$

The pair has only one independent coordinate. Its projective odds recover a scalar balance coordinate with balanced value one. This gives a direct bridge to carrier-level scalar order without identifying the internal pair with any particular measured order variable.

$$\begin{aligned} r_{\{U,n+1\}} &:= \omega_{\{n+1\}}^{\wedge\{\text{keep}\}} / \omega_{\{n+1\}}^{\wedge\{\uparrow\}} \\ &= \Lambda_n^{\wedge\{\uparrow\}} / \Lambda_n^{\wedge\{\text{keep}\}}, \quad \theta_{\{U,n+1\}} := \log \\ &\quad r_{\{U,n+1\}}, \\ \omega_{\{n+1\}}^{\wedge\{\text{keep}\}} &= (1 + e^{-\theta_{\{U,n+1\}}})^{-1}, \\ \omega_{\{n+1\}}^{\wedge\{\uparrow\}} &= 1 - \omega_{\{n+1\}}^{\wedge\{\text{keep}\}} \end{aligned} \tag{34}$$

At $\Lambda^{\wedge\{\text{keep}\}} = \Lambda^{\wedge\{\uparrow\}}$, the unique state is $(1/2, 1/2)$ and $r_U = 1$. If one load grows relative to the other, its own order weight decreases while the opposite weight increases. The mechanism is therefore reciprocal negative feedback. A branch with zero closure load cannot support a genuine dual reclosure and is mapped to the terminal case by the universal gate below.

4.5 Cross-fed packet and exact non-expansion

$$\begin{aligned} \mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}} &:= \sqrt{(\omega_{\{n+1\}}^{\wedge\{\text{keep}\}}) \mathcal{D}_n^{\wedge\{\text{keep}\}} \oplus \sqrt{(\omega_{\{n+1\}}^{\wedge\{\uparrow\}}) \mathcal{D}_n^{\wedge\{\uparrow\}}}} \end{aligned} \quad (35)$$

$$\begin{aligned} \|\mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}}\|_{\mathcal{G}_n \oplus \mathcal{G}_n}^2 &= 2\Lambda_n^{\wedge\{\text{keep}\}} \\ &\quad \Lambda_n^{\wedge\{\uparrow\}} / S_n \\ &\leq S_n / 2 \leq \|\mathcal{D}_n\|_{\mathcal{G}_n}^2 / 2 \end{aligned} \quad (36)$$

Equation (36) is an exact local theorem, not a simulation result. C_N and B_N are commuting positive contractions with $C_N + B_N = I$, the witness projection is $\mathcal{G}_n$ -orthogonal, and the reciprocal weights solve Equation (32). Consequently the dual-order coupler cannot manufacture norm gain inside the structural master metric. This directly excludes an interpretation of endogenous order as a hidden amplification source.

5. Canonical Endogenous Dual-Order Strong Universal Reclosure Architecture

5.1 Canonical cross-fed analysis operator

The remaining potential freedom is next-level synthesis. Instead of postulating a map J_n , the Canonical Closure Edition first defines the analysis operator generated by the already fixed neutral pair, persistent projector, and reciprocal order weights.

$$\begin{aligned} \mathcal{A}_n^{\wedge\{\text{DO}\}} &:= [\sqrt{(\omega_{\{n+1\}}^{\wedge\{\text{keep}\}}) \mathcal{P}_{\{\mathcal{W}_n\}}^{\wedge\{\text{G}\}} C_N(Y_n) ; } \\ &\quad \sqrt{(\omega_{\{n+1\}}^{\wedge\{\uparrow\}}) \mathcal{P}_{\{\mathcal{W}_n\}}^{\wedge\{\text{G}\}}} \\ B_N(Y_n)], \quad \mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}} &= \mathcal{A}_n^{\wedge\{\text{DO}\}} \mathcal{D}_n \end{aligned} \quad (37)$$

Its polar decomposition determines a unique partial isometry on the active support. Let $Q_n = (\mathcal{A}_n^{\wedge\{\text{DO}\}} \mathcal{A}_n^{\wedge\{\text{DO}\}})^{\wedge\{1/2\}}$. The Moore–Penrose inverse is used only on $\text{supp}(Q_n)$. The resulting J_n is fixed by X_n and satisfies $J_n^\dagger J_n = P_{\text{supp}(Q_n)}$; on the active support it is an isometry. No target label or free transport matrix remains.

$$\begin{aligned} Q_n &:= (\mathcal{A}_n^{\wedge\{\text{DO}\}} \mathcal{A}_n^{\wedge\{\text{DO}\}})^{\wedge\{1/2\}}, \\ J_n &:= \mathcal{A}_n^{\wedge\{\text{DO}\}} Q_n^{\wedge\{+\}}, \\ J_n^\dagger J_n &= P_{\{\text{supp}(Q_n)\}} \end{aligned} \quad (38)$$

5.2 Canonical capacity transport and positive rebuild

Every self-adjoint capacity operator is transported by compression through the same polar partial isometry. The new active carrier space is the closed range of J_n inside the direct-sum space, so the transported capacities and the cross-fed packet live on the same mathematical object. Positivity, where present, and self-adjointness are preserved by compression. The support-coordinate expression $J_n^\dagger (\mathfrak{A}_n \oplus \mathfrak{A}_n) J_n$ is unitarily equivalent to the range compression. The next formation operator is therefore a derived object rather than a stage-dependent choice.

$$\begin{aligned} \mathcal{H}_{\{n+1\}} &:= \text{cl Ran}(J_n), \quad \mathcal{P}_{\{n+1\}} := J_n J_n^\dagger, \\ \mathfrak{A}_{\{n+1\}} &:= \mathcal{P}_{\{n+1\}} (\mathfrak{A}_n \oplus \mathfrak{A}_n) \mathcal{P}_{\{n+1\}} \\ \mathcal{H}_{\{n+1\}} &\simeq J_n^\dagger (\mathfrak{A}_n \oplus \mathfrak{A}_n) J_n, \\ \mathfrak{A} &\in \{\mathcal{D}, c, N\}, \\ K_{\{n+1\}} &:= c_{\{n+1\}} + N_{\{n+1\}} - \mathcal{D}_{\{n+1\}} \end{aligned} \quad (39)$$

The cross-fed packet then enters the same positive Gram reconstruction used by the universal closure architecture. Positivity is regenerated, not assumed inductively.

$$\begin{aligned} H_{\{n+1\}} &:= B_C[\mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}}]^\dagger W_N[\mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}}] B_C[\mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}}] \geq 0, \\ Y_{\{n+1\}} &:= L_C[\mathcal{D}_{\{n+1\}}^{\wedge\{\text{DO}\}}]^\dagger H_{\{n+1\}} \end{aligned} \quad (40)$$

$$L_C[\mathcal{D}_{\{n+1\}}^{\{\mathcal{D}\}}] \geq 0$$

5.3 Universal gate, terminality, and the EDO-URE

A nonterminal branch must possess both persistent poles, an isolated witness gap, and the same simple-negative-mode formation geometry used at Level-0. Failure is never repaired by adding a sector-specific fallback rule.

$$g_U^{\mathcal{D}\{n\}} := \mathbb{1}_{\{ \{ \Lambda_n^{\text{keep}} \} \Lambda_n^{\uparrow} > 0, \quad (41)$$

$$\gamma_n > 0, \lambda_{\min}(K_{\{n+1\}}) < 0, \\ \text{mult}(\lambda_{\min}(K_{\{n+1\}})) = 1, \Delta_{\{n+1\}} > 0 \}$$

$$X_{\{n+1\}} = \mathbb{U}[X_n] := \{ C_N(Y_{\{n+1\}}) \mathcal{D}_{\{n+1\}}^{\mathcal{D}\{n\}} \quad (42)$$

$$\wedge^{\mathcal{D}\{n\}}, \text{ if } g_U^{\mathcal{D}\{n\}} = 1; \\ \perp, \quad \text{if } g_U^{\mathcal{D}\{n\}} \\ (n) = 0 \}$$

$$\mathbb{U}: X_{\text{adm}} \rightarrow X_{\text{adm}}, \quad (43)$$

$$X_{\text{adm}} := X_{\text{adm}} \cup \{\perp\}, \quad \mathbb{U}(\perp) = \perp,$$

$$X_n = \mathbb{U}^n[X_0], \quad \omega_{\{U, n+1\}} \\ = \mathcal{O}_U[\mathbb{U}^n(X_0)]$$

5.4 Carrier order is a readout, not a generator input

The earlier scalar order relation remains useful at the empirical interface. Its status is now unambiguous: a carrier-specific scalar may be compared prospectively with the internal projective coordinate r_U , but it does not enter $\mathbb{U}$. A successful carrier theory must specify this readout before outcome inspection.

$$\varphi_{\text{obs}} \nu(\varphi_{\text{obs}}) = \Phi_{\text{src}}, \quad \varphi_{\text{obs}} \approx \quad (44)$$

$$\mathcal{R}_{\varphi}[r_U(X)] \quad (\text{carrier-level realization only})$$

5.5 Hilbert-space closure beyond finite matrices

The universal resolvent core does not depend on finite matrix dimension. Let K be a closed densely defined operator on a separable Hilbert space. Then $Y = K^\dagger K$ is nonnegative self-adjoint, so the spectral theorem gives bounded positive resolvents even when Y itself is unbounded [9,10]. This supplies the natural operator-theoretic extension needed for continuum carrier models without changing the neutral equations.

$$Y := K^\dagger K \geq 0 \text{ self-adjoint}, \quad C_N(Y) := (I + Y)^{-1}, \quad (45)$$

$$B_N(Y) := I - C_N(Y), \\ 0 \leq C_N(Y) \leq I, \quad 0 \leq B_N(Y) \leq I$$

The corresponding positive master geometry is the quadratic-form graph space. All load and metric-projection statements are interpreted on its closed form domain. The finite-dimensional theory is the bounded special case.

$$\mathcal{Q}(Y) := \text{Dom}(Y^{\{1/2\}}), \quad \|z\|_G^2 := \|z\|^2 + \|Y^{\{1/2\}} z\|^2, \quad (46)$$

G-form-closed witness subspaces admit unique
orthogonal projections

5.6 Canonicity and no-hidden-retuning theorem

Conditional theorem. Under the declared Level-0 formation assumptions, exact neutral elimination, the canonical persistence definition in Equation (27), positive spectral isolation, reciprocal balance in Equation (32), polar support transport in Equation (38), positive Gram reconstruction, and the common formation gate, every generator-level object is fixed by X_n and universal rules. $\mathbb{U}$ is therefore a totalized self-map on the declared admissible operator category. The theorem requires no stage predicate, externally supplied order coordinate, tunable persistence window, or freely

chosen next-level transport operator. Natural-system realization is deliberately not inserted into this theorem; it is tested downstream.

Figure 1. Canonical Endogenous Dual-Order Strong Universal Reclosure architecture. The order pair is uniquely fixed by reciprocal balance, the witness subspace is defined by bounded non-decay, and next-level transport is the polar partial isometry of the cross-fed analysis operator. The same engine rebuilds positive geometry and either returns the next admissible state or terminates at $\perp$.

6. Target-independent invariant classes and natural realization

The Endogenous Dual-Order Strong Universal Form removes named stage predicates from the mathematics. After $\mathbb{U}$ generates an iterate, that iterate is represented only by a universal invariant signature containing spectral, persistence, endogenous dual-order, factorization, return, and nesting data. Two states belong to the same mathematical class only when their invariant structures agree under an admissible structure-preserving isomorphism.

$$Q_U := \bar{X}_{\text{adm}} / \sim_U \quad (47)$$

$$\begin{aligned} \Pi_s: N_s &\rightarrow Q_U, \\ \varepsilon_s(x) &:= d_U(\Pi_s(T_s x), [\mathbb{U}(X_x)]_{\sim_U}) \end{aligned} \quad (48)$$

Natural names enter only through realization maps Π_s from independently described empirical systems into the universal quotient. A proposed atom, molecule, protocell, organism, conscious state, or qualitative content is therefore not generated because a predicate demanded that label. It is supported only if prospectively measured carrier data map into an already-existing invariant class and if observed transitions commute with $\mathbb{U}$ within the frozen error metric.

7. Physical closure from the same architecture

7.1 Local Physical Closure Principle

The Local Physical Closure Principle (LPCP) states that once an admissible closure enters a smooth, reversible, locally geometric regime, local propagation is the continuum limit of covariantly transported neighboring closure differences, using the same positive neutral stiffness that controls admissibility [1]. LPCP is a physical principle, not a Level-0 theorem.

$$\Delta_{\{\varepsilon, v\}} q(x) = U_C(x, x + \varepsilon v) q(x + \varepsilon v) - q(x) \quad (49)$$

$$\Delta_{\{\varepsilon, v\}} q / \varepsilon = \nabla_v \wedge C q + O(\varepsilon) \quad (50)$$

$$Z_C = H_C^{\{1/2\}} C_N(Y_C[q_*]) H_C^{\{1/2\}} > 0 \quad (51)$$

$$S_{\text{LPC}}^{\{(2)\}}[q] = \mathcal{K}_0 \int d^4x \sqrt{-g} [-\frac{1}{2} g^{\{\mu\nu\}} \langle \nabla_\mu \wedge C q, Z_C \nabla_\nu \wedge C q \rangle - V_C^{\{(2)\}}(q)] \quad (52)$$

$$V_C(q) = \frac{1}{2} \langle D_C(q), C_N(Y_C(q)) D_C(q) \rangle \quad (53)$$

The local response Hessian follows from derivatives of the same resolvent, rather than from an independent sector response.

$$\partial_A C = -C Y_{,A} C \quad (54)$$

$$\partial_A \partial_B C = C Y_{,A} C Y_{,B} C + C Y_{,B} C Y_{,A} C - C Y_{,AB} C \quad (55)$$

In the minimal single-metric, two-derivative branch, the principal symbol of every nondegenerate fundamental massless block is proportional to one metric null form. Under additional standard assumptions, the geometric gauge block is related to Fierz–Pauli massless spin-2 and the Einstein infrared branch [13–17]. These are conditional physical consequences, not additional BFG fundamental laws.

$$\sigma_2(\Gamma_a)(x, k) = Z_a g^{\{\mu\nu\}} k_\mu k_\nu P_a \quad (56)$$

$$g^{\{\mu\nu\}}_{\mu\nu} = 0 \quad (57)$$

7.2 Physical realization map

A natural system is associated with the local physical branch only after the universal invariant class has been generated. The empirical realization map tests whether a smooth carrier, nondegenerate soldering, positive closure stiffness, reversible local transport, stable characteristic structure, and a carrier order readout compatible with r_U or ω_U correspond to that class under the Local Physical Closure Principle. These carrier conditions do not alter $\mathbb{U}$.

Physical-stage

status

The existence of an admissible local geometric branch is conditional. The identification of the resulting metric and spectrum with observed spacetime and gravity remains empirical. The standard field-theory literature constrains the carrier interpretation [13–17]; it does not independently derive BFG.

8. Matter: nested nuclear, atomic, and molecular closure

Atomic and molecular systems are treated as realization tests of already-generated invariant classes, not as definitions of those classes. Established quantum mechanics, electrodynamics, nuclear physics, and chemistry supply the carrier observables [18]. BFG contributes the universal retained/complement, persistence, nesting, and export coordinates against which those observables are compared.

8.1 Atomic realization test

An atomic realization is supported only when an independently measured bound system maps to a universal class with an isolated persistent spectral block, noncollapsed distinguishability, bounded recursive transport, admissible complement/export, and an internally generated dual-order state compatible with the carrier data. The class exists mathematically without the word “atom”; the label is attached only after the empirical map succeeds. No atomic Hamiltonian, threshold, or order parameter is inserted into $\mathbb{U}$.

$$\Pi_{\text{atom}}: N_{\text{atom}} \rightarrow Q_U \text{ is accepted only from independently measured carrier data} \quad (58)$$

This is intentionally a structural implication rather than a derivation of the hydrogen spectrum from a scalar toy load. The no-retuning claim is that the BFG closure law is unchanged; standard physics supplies the carrier Hamiltonian and measured spectral data.

8.2 Molecular and chemical reclosure

A molecular realization is tested by whether independently identified lower bound systems map jointly into a higher universal class whose retained response is nonfactorizable while the lower subclosures remain recoverable. The higher class is generated by $\mathbb{U}$ before chemical naming; chemistry supplies the carrier Hamiltonian and measured relational spectrum.

$$C_{\text{mol}} \neq C_{\text{atom},1} \oplus C_{\text{atom},2} \quad \text{when the higher retained response is nonfactorizable} \quad (59)$$

The distinction between constitutive nesting and temporal generation is important. The hierarchy expresses closure precedence. It does not imply that every molecule must literally replay Level-0 or atomic formation as a temporal developmental sequence.

9. From chemistry to abiogenesis

The molecular-to-biological transition is treated as a falsifiable generativity problem rather than an adjacent pair of labels [5]. External origin-of-life research shows that several carrier-level ingredients are experimentally tractable: activated pyrimidine ribonucleotides can arise under prebiotically plausible chemistry [19], common precursor chemistry can feed RNA, amino-acid, and lipid building blocks [20], template-directed polymer synthesis can occur in

model protocells [21,22], protocell populations can exhibit competition [23], and recent autocatalytic systems can couple reaction dynamics to compartment growth and reproduction-like behavior [24]. These results constrain carrier plausibility; none validates BFG.

9.1 Prebiotic realization test

Reaction-network organization, compartment persistence, throughput, renewal, heredity, and repeated-cycle persistence are now treated strictly as empirical coordinates used to map prebiotic systems into the universal quotient. They are not terms in the generator and do not define a prebiotic mathematical class in advance. The only mathematical generator remains Equation (42).

$$\Pi_{\text{prebio}}: N_{\text{prebio}} \rightarrow Q_U \quad (60)$$

$$\varepsilon_{\text{cycle}}(x) = d_U(\Pi_{\text{prebio}}(T_{\text{cycle}} x), [\mathbb{U}(X_x)]_{\sim U}) \quad (61)$$

For empirical work, the multicomponent observation vector $q_B, q_F, q_R, q_H, q_C, q_S$ remains useful [5], but its role is diagnostic. Those observables are measured independently and compared with invariant coordinates of $\mathbb{U}$ -generated classes. No component is permitted to change the universal capacity transport, neutral response, positive-metric witness rule, endogenous dual-order mapping, persistence rule, or gate.

$$\vec{q}_{\text{abi}} = (q_B, q_F, q_R, q_H, q_C, q_S) \quad (62)$$

A strong empirical test compares the full frozen BFG readout against capacity-matched hierarchy-only, compartment-only, heredity-only, and shuffled-coupling nulls. The repeated-cycle requirement remains: a one-off vesicle or isolated copying event is not enough to identify biological closure [5,25].

10. Cellular, multicellular, and organismic reclosure

10.1 Cellular closure

The previously described biological ordering is retained only as a natural-history and systems-biology comparison sequence [26]. In the Strong Universal mathematics there is no cellular, tissue, organ, or organism generator. There is only the orbit of $\mathbb{U}$ in the target-independent quotient; biological labels are empirical realization assignments made after the invariant classes exist.

A cellular realization therefore requires prospectively measured boundary persistence, throughput, informational renewal, and recursive self-maintenance across turnover to co-localize in one generated invariant class. Failure of that mapping is a failure of the proposed biological realization, not a reason to modify the universal operator.

10.2 Multicellular projection and morphogenesis

A prior organismic formulation treats local cellular closures as projecting into a form/field domain, being read out, differentiated, and reintegrated into a global closure [27]. Classical developmental biology provides independent mechanisms—reaction–diffusion, positional information, morphogen interpretation, gene-regulatory organization, mechanics, and bioelectric pattern control [28–32]. BFG does not replace these mechanisms. The unification claim is that they can be evaluated as carrier realizations of a common closure transition.

$$\Pi_{\text{org}}: N_{\text{org}} \rightarrow Q_U \quad (63)$$

A multicellular mass does not qualify merely because cell number increases. The organismic predicate requires persistent global integration, preserved local differentiation, admissible projection/readout, regulated export, and failure under the appropriate ablations. The distinction cell growth $\neq$ organism formation is retained [27].

10.3 Organismic closure

Organismic realization is treated as a many-subclosure empirical mapping problem. Local biological closures must remain individually recoverable while a higher nonfactorizable invariant class captures global persistence, differentiation, return, and regulated export. The organism label is assigned only after this higher class has been generated and matched; it never enters the formation rule.

11. Sensorimotor, cognitive, and memory closure

Sensorimotor, learning, memory, self-model, perspective, and identity concepts are likewise downstream realization hypotheses [26]. The universal mathematics instead tracks target-independent quantities such as externally driven distinguishability, cross-channel nonfactorization, recursive return, witness persistence, and self-referential reachability. Only after these coordinates are estimated independently may cognitive labels be attached.

Table 2. Target-independent cognitive invariant coordinates used only for downstream realization tests.

Invariant coordinate	Target-independent meaning
External distinguishability	Externally driven differences remain recoverable after neutral resolution and recursive transport.
Nonfactorization	Cross-channel closure cannot be reduced to a direct sum of independent retained blocks.
Return depth	Perturbations re-enter the same closure through a bounded sequence of return mappings.
Witness persistence	A prospectively selected invariant subspace remains power bounded across separated recursions.
Adaptive plasticity	Transverse degrees of freedom change while a declared witness remains recoverable.
Self-reference reachability	A closure representation can reach and be reached from a representation of its own persistent state.
World/self separation	Two distinguishable relational sectors remain jointly mediated without collapse.
Integration margin	Shared-neutral off-diagonal coupling remains nonzero under preserved differentiation.
Differentiation margin	The active retained operator keeps a positive minimum singular-value margin.

12. Thermodynamic maintenance as a readout, not a second consciousness law

A prior thermodynamic model introduced explicit entropy, information, and energy-balance equations [35]. In the present unification those equations are not used as a second fundamental definition of consciousness. Instead, thermodynamic quantities are read from a physical carrier that realizes the universal retained/complement partition.

For a normalized carrier state D , define retained and complementary quadratic fractions:

$$r_N(D) = \langle D, C_N(Y)D \rangle / \langle D, D \rangle \quad (64)$$

$$b_N(D) = \langle D, B_N(Y)D \rangle / \langle D, D \rangle = 1 - r_N(D) \quad (65)$$

A physical model may then map these fractions into bound energy, exported heat, chemical work, or information-processing costs. Landauer's result constrains the energetic cost of logically irreversible information operations [36], but it does not imply that consciousness is simply entropy reduction.

$$E_{\text{bind}} = \mathcal{R}_E^{\{\text{bind}\}}[r_N, \text{carrier}] \quad (66)$$

$$E_{\text{exp}} = \mathcal{R}_E^{\{\text{out}\}}[b_N, \text{carrier}] \quad (67)$$

The readout maps $\mathcal{R}_E$ are empirical constitutive maps, not new fundamental BFG dynamics. They do not enter $\mathbb{U}$ as external energetic controls. The universal map feeds back only the structural state that it generates and its endogenous dual-order readout. A measured carrier may change the represented X and therefore Y through the already declared upstream map, but no domain-specific thermodynamic coefficient is allowed to retune the EDO-URE.

13. Conscious subjectivity as a shared-neutral EDO-URE realization class

The shared-neutral cognitive formulation supplies a positive-load/resolvent family compatible with the universal architecture [12]. In the Endogenous Dual-Order Strong Universal Form, however, “conscious subjectivity” is not a mathematical stage predicate and the local scalar φ used in the carrier model is not a fundamental universal input. It is an empirical realization coordinate to be compared with a declared readout of r_U or ω_U . A measured cognitive system must map into an invariant class exhibiting simultaneous shared-neutral integration, preserved differentiation, bounded recursion, self-indexing, and identity persistence under fixed rules.

$$Y_\varphi = M_\varphi^\dagger M_\varphi \geq 0 \quad (68)$$

$$C_\varphi = (I + Y_\varphi)^{-1} \quad (69)$$

$$Q_\varphi = Y_\varphi (I + Y_\varphi)^{-1} = I - C_\varphi \quad (70)$$

Relative to a fixed channel decomposition with projectors P_i , exact channel factorization is equivalent for Y_φ and C_φ . Shared-neutral integration therefore requires nonzero off-diagonal response, while differentiation is protected by the smallest singular value.

$$Y_{\{ij\}}(\varphi) = P_i Y_\varphi P_j, \quad C_{\{ij\}}(\varphi) = P_i C_\varphi P_j \quad (71)$$

$$\Xi_\varphi = [\Sigma_{\{i \neq j\}} \|P_i C_\varphi P_j\|_{\text{HS}}^2] / \|C_\varphi\|_{\text{HS}}^2 \quad (72)$$

$$\Delta_\varphi = \sigma_{\min}(C_\varphi|_{\{H_{\text{act}}\}}) = 1/[1 + \lambda_{\max}(Y_\varphi|_{\{H_{\text{act}}\}})] \quad (73)$$

World/self relation, self-indexing, identity persistence, and bounded recursion therefore remain jointly testable coordinates, not definitional labels. The empirical identification of a matching invariant class with conscious subjectivity remains a hypothesis [12].

$$K_\varphi = R_\varphi C_\varphi \quad (74)$$

$$K_\varphi^\dagger L K_\varphi \leq L \quad (75)$$

$$\sup_n \|K_\varphi^n x\|_L < \infty \quad (76)$$

Perturbational-complexity and near-criticality research provide external benchmarks for integration-plus-differentiation and bounded dynamical regimes [42,43]. BFG does not identify its variables directly with PCI or with a generic criticality index; such mappings must be preregistered.

14. Phenomenal content and qualitative identity

Qualitative content is handled in the same order: first the universal equivalence class, then the phenomenological realization claim [44]. The formal object $Q[X]$ is a closure-equivalence class determined by invariant structure. Whether one such class corresponds to experienced redness, pain, or another quale is not built into the quotient and must be established independently.

14.1 BFG closure equivalence

Let $\mathcal{G}_{\text{adm}}$ be the set of admissible transformations that preserve the declared witness spectrum, the neutral-response spectra, the local order state within tolerance, and the restricted recursive transport. Define:

$$X \sim_{\text{BFG}} X' \Leftrightarrow \exists G \in \mathcal{G}_{\text{adm}}: X' = GX \text{ and } \mathbb{I}_U(X') = \mathbb{I}_U(X) \quad (77)$$

$$\mathbb{I}_U(X) := \{\text{Spec } C_N(Y_X), \text{Spec } B_N(Y_X), \omega_U(X), r_U(X), \mathcal{W}_X, R_C|_{\{\mathcal{W}_X\}}\} \quad (78)$$

$$Q[X] := [X]_{\sim_{\text{BFG}}} \quad (79)$$

The formal qualitative identity is therefore an equivalence class of the closure geometry, not an extra postulated substance. For two content closures:

$$Q_{\text{red}} \neq Q_{\text{blue}} \Leftrightarrow [C_{\text{red}}]_{\sim_{\text{BFG}}} \neq [C_{\text{blue}}]_{\sim_{\text{BFG}}} \quad (80)$$

This construction closes the formal operator gap only at the mathematical level. The empirical claim that a particular equivalence class is phenomenal redness still requires content-specific neural and behavioral tests. If closure invariants fail to distinguish red from blue after controlling local activation, or if the equivalence relation can only be defined using the target report itself, the BFG qualia claim is weakened or fails [44].

14.2 Ownership and selfhood

Ownership is likewise formulated as a relation between a content-equivalence class and a persistent identity witness after both have been generated within the universal quotient. The phrase “my experience” is therefore an empirical interpretation of a bidirectional closure relation, not a primitive term in $\mathbb{U}$.

$$\text{SelfOwned}(Q_i) = 1 \Leftrightarrow Q_i \text{ is bidirectionally reachable from the persistent identity witness } \mathcal{W}_{\text{id}} \quad (81)$$

External work on bodily self-consciousness supports the broader empirical fact that ownership, self-location, perspective, and multisensory self-processing can dissociate [45]. It does not validate the BFG selfhood operator; it supplies independent phenomena against which the closure architecture can be tested.

15. The complete cosmic precedence map

The cosmic precedence map is retained only as an empirical comparison scaffold. It does not define the order in which $\mathbb{U}$ must produce named natural categories. The mathematical orbit contains unlabeled invariant classes; physical, chemical, biological, cognitive, and phenomenal names are assigned afterward if independent carrier data realize them. Structural precedence therefore remains conceptually useful without becoming a hidden stage predicate.

Table 3. Natural-system realization scaffold mapped onto target-independent EDO-URE invariant coordinates.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
0	Level-0 admissibility	Difference/coherence/neutral capacities and an isolated negative formation mode.
1	Primitive formed structure	Observed carrier candidate maps to a class with a realized Level-0 branch and finite positive load.
2	Neutral-resolved structure	Observed response is consistent with the exact C_N/B_N partition.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
3	Persistent structure	Measured dynamics support a prospectively isolated power-bounded witness block.
4	Local physical realization	Independent carrier data satisfy the physical realization conditions associated with a generated class.
5	Bound physical realization	Measured bound structure maps to a spectrally isolated retained class with admissible export.
6	Nuclear/atomic realization	Quantum carrier data map to a nested bound class with persistent spectral separation.
7	Molecular realization	Chemical data map multiple lower bound realizations into a higher nonfactorizable class.
8	Macromolecular / sequence realization	Measured relational ordering persists across reclosure-relevant perturbations.
9	Informational realization	Alternative sequence states remain discriminable and transportable under fixed carrier mapping.
10	Replication realization	A closure-relevant witness is reproduced into an additional same-level realization.
11	Error-correction realization	Perturbed carrier states return to the same target-independent invariant class.
12	Metabolic throughput realization	Persistent retention coexists with regulated measured export and renewal.
13	Boundary realization	Measured inside/outside partition remains stable while selective exchange persists.
14	Protocellular realization	Boundary, throughput, renewal, heredity, coupling, and repeated-cycle persistence co-map to one class.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
15	Cellular realization	A protocellular carrier realization remains recursively self-maintaining across material turnover.
16	Projection realization	Measured persistent complementary structure becomes accessible to another local realization.
17	Readout realization	Projected structure measurably changes another closure-mediated carrier state.
18	Feedback realization	Projection and return form a bounded empirically observed recursive loop.
19	Network / field realization	Multiple local realizations map jointly into a nonfactorizable higher class.
20	Attractor realization	Perturbed carrier states return to one invariant basin under fixed measurement rules.
21	Differentiation realization	Distinct local basins persist while co-mapping to one higher class.
22	Tissue realization	Many cellular realizations support a persistent spatial-functional higher mapping.
23	Organ realization	Specialized tissue realizations map into a higher persistent functional class.
24	Organ-system realization	Multiple organ realizations jointly map into an integrated higher class.
25	Organism realization	Global empirical organization persists despite internal turnover and differentiation.
26	Sensory realization	External perturbations map to discriminable internal retained coordinates.
27	Integration realization	Multiple channels map to a shared nonfactorizable neutral-load class.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
28	Valence realization	Alternatives exhibit reproducibly distinct persistence/export gradients.
29	Action realization	Measured export changes the exterior and returns as structured input.
30	Learning realization	Future measured transition structure changes after prior perturbation/feedback.
31	Memory realization	A witness remains recoverable across temporally separated cycles.
32	Self-model realization	Carrier data support a persistent internal representation of the system's own state.
33	Perspective realization	World-related and self-related coordinates remain distinct but jointly mediated.
34	Identity realization	A persistent self-index remains while transverse carrier degrees adapt.
35	Conscious-subjectivity realization hypothesis	Independent cognitive data map jointly to integration, differentiation, self/world structure, identity, and bounded recursion.
36	Phenomenal-unity realization hypothesis	Multiple content-related realizations occupy one globally connected differentiated class.
37	Qualitative-content realization hypothesis	Content identity is tested against a target-independent closure-equivalence class.
38	Self-owned experience realization hypothesis	Content-related class is bidirectionally linked to an independently estimated identity witness.
39	Narrative / intersubjective realization hypothesis	Temporally extended self-model and other-subject relations are tested only after class generation.

Index	Natural realization hypothesis	Independent carrier signature to be mapped into Q_U
40	Knowledge / science / civilization realization hypothesis	Collective carrier systems are mapped post hoc into higher invariant classes; highly speculative empirical extension.

16. Executed numerical demonstrations

All simulations in this section were executed for the present whitepaper in Python 3.13.5 using NumPy 2.3.5 and Matplotlib 3.10.8. They are deterministic and dimensionless. No empirical dataset was fitted. The simulations are designed to test algebraic or stability consequences of the declared equations and to expose failure surfaces; they do not demonstrate that any physical, biological, or conscious system realizes BFG.

16.1 Exact neutral-partition surface

For two independent eigenloads $y_1, y_2 \geq 0$, define $c_i = (1 + y_i)^{-1}$ and $b_i = y_i / (1 + y_i)$. The exact partition $c_i + b_i = 1$ holds independently on each eigenmode. A symmetric diagnostic of simultaneous retained/complement availability is:

$$\Gamma_{rc}(y_1, y_2) = \frac{1}{2}[4c_1b_1 + 4c_2b_2], \quad 0 \leq \Gamma_{rc} \leq 1 \quad (82)$$

The numerical sweep used $y_1, y_2 \in [10^{-2}, 10^2]$. The maximum grid value was 0.999790 at $y_1 \approx 0.9715$, $y_2 \approx 0.9715$; the maximum absolute partition error was $2.220e-16$, consistent with floating-point roundoff. The analytic maximum $\Gamma_{rc} = 1$ occurs at $y_1 = y_2 = 1$, where retained and complementary fractions are balanced mode by mode.

Figure 2. Executed 3D neutral-partition simulation. The surface is derived solely from the exact BFG resolvent pair. The ridge collapses to the analytic balanced maximum at $y_1 = y_2 = 1$; no sector-specific response function is used.

16.2 Spectral generativity normal form

The spectral gate depends only on the sign of the next-level minimum formation eigenvalue. To visualize this sign change without assigning a physical sector, a two-mode symmetric normal form was used:

$$K_{nf}(\delta, g) = [[1 + \delta, -g], [-g, 1 - \delta]] \quad (83)$$

$$\lambda_{\min}(K_{nf}) = 1 - \sqrt{(\delta^2 + g^2)} \quad (84)$$

$$g_U = 1 \Leftrightarrow \delta^2 + g^2 > 1 \quad (85)$$

A dense sweep over $\delta \in [-1.6, 1.6]$ and $g \in [0, 1.6]$ reproduced the analytic circular sign boundary exactly to grid resolution. 69.5% of the displayed rectangular parameter window lies in the negative-eigenvalue region; this percentage has no physical meaning and depends only on the chosen plotting window. The scientific content is the visible gate surface $\lambda_{\min} = 0$.

Figure 3. Executed 3D spectral-gate normal form. The surface illustrates the canonical rule $g_U = 1$ when the candidate next-level formation operator acquires a negative minimum eigenvalue. The matrix is a pedagogical normal form, not a carrier law.

16.3 Order–feedback stability corridor

The declared order-indexed toy Jacobian uses $v(\varphi) = \exp[-4(\varphi - 1)^2]$, $\alpha = 0.35$, $\beta = 0.25$, $d = 0.45$, and $c = 0.08$ [5]. The present work re-executed the full (φ, B) spectral-radius sweep.

$$v(\varphi) = \exp[-4(\varphi - 1)^2] \quad (86)$$

$$J(\varphi, B) = [[B + 0.35(1 - \nu), 0.08\nu], [0.08\nu, 0.45 + 0.25(1 - \nu)]] \quad (87)$$

At $B=0.82$ the executed values are $\rho(J) = 0.8366$ at $\varphi=1.0$ and 0.9897 at $\varphi=1.4$. At $B=0.90$ and $\varphi=1.4$, $\rho(J) = 1.0690$, crossing the linear stability boundary. Across the displayed $\varphi \in [0.45, 1.55]$, $B \in [0.55, 0.98]$ grid, 78.9% of sampled points had $\rho < 1$. That fraction is a plotting statistic, not a universal prediction.

Figure 4. Executed 3D spectral-radius surface for the order-indexed interface toy model. The surface reproduces the finite joint stability corridor associated with the cited parameterization [5]. Linear spectral stability is necessary only for this fixed linear toy; nonnormal and time-varying systems require stronger power-bounded or Lyapunov analysis.

16.4 Shared-neutral cognitive integration

The five-channel cognitive simulation was re-executed from the previously defined matrices [12]. The mediation map uses $D = \text{diag}(0.55, 0.65, 0.75, 0.85, 0.95)$, a symmetric five-node ring with three weak chords normalized to spectral norm one, and $M_\varphi = D + g\varphi W$. With $\Phi_{\text{src}} = 1$ and $\nu(\varphi) = 1 + 0.45\varphi$, the positive order fixed point is:

$$\varphi^* = [-1 + \sqrt{(1 + 4a\Phi_{\text{src}})}] / (2a) \quad (88)$$

The executed value is $\varphi^* = 0.7481$. At $g = 0, 0.4, 0.8, 1.2, 1.6$ the integration/differentiation pairs are respectively 0.0: (0.0000, 0.5256); 0.4: (0.0261, 0.4482); 0.8: (0.0819, 0.3442); 1.2: (0.1350, 0.2642); 1.6: (0.1697, 0.2061). These reproduce the previously defined trade-off [12]: coupling raises nonfactorizable integration while lowering the guaranteed differentiation margin.

Figure 5. Executed 3D shared-neutral integration surface Ξ_φ . Integration vanishes in the $g=0$ ablation and grows as the common mediation map develops off-diagonal channel coupling. The plot is an internal mathematical demonstrator, not a consciousness measurement.

Figure 6. Executed 3D differentiation surface $\Delta_\varphi = \sigma_{\min}(C_\varphi)$ over the same parameter plane. Increasing shared load can erode the minimum distinguishability margin; this is why BFG does not identify maximal coupling with consciousness.

16.5 Recursive critical surface and derivative identity

At the benchmark $\varphi^* = 0.7481$ and $g = 0.8$, the previous parameterization [12] uses $R_0 = 1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$. The present execution gives:

$$K_\varphi(r, g) = rR_0C_\varphi(g) \quad (89)$$

$\rho(R_0C_\varphi) = 0.8711$, so $r_{\text{crit}} = 1/\rho = 1.1480$. Choosing $r = 1.1422$ yields $\rho(K_\varphi) = 0.995000$. The analytic resolvent derivative $\partial_\varphi C_\varphi = -C_\varphi Y'_{\varphi} C_\varphi$ was checked against a centered $h = 10^{-6}$ finite difference with relative Frobenius error $4.297e-10$.

Figure 7. Executed 3D recursive stability surface for $K_\varphi = rR_0C_\varphi$ at fixed $g = 0.8$. The critical level $\rho = 1$ separates spectrally subcritical and supercritical regions for this fixed linear model.

16.6 Bounded near-critical trajectory

Using the near-critical gain above and the deterministic initial state $s_0 = (0.90, -0.55, 0.35, 0.25, -0.20)$, 160 iterations remained bounded. The initial Euclidean norm was 1.1565, the maximum norm over the run was 1.3535, and the final norm was 0.6575. The trajectory is shown in the first three principal coordinates of the 161 simulated states.

Figure 8. Executed bounded near-critical trajectory projected into three principal coordinates. The run demonstrates power-bounded recurrent behavior for the chosen fixed linear model. It does not demonstrate phenomenal consciousness.

16.7 Endogenous dual-order cross-feeding surface

The reciprocal order law in Equation (33) was evaluated directly over four decades of positive keep and upward structural load. Because the two components are cross-fed and normalized, the surface is scale invariant: only the ratio $\Lambda^\uparrow/\Lambda^{\text{keep}}$ determines the order coordinate. At equal loads, $\omega_U^{\text{keep}}=\omega_U^\uparrow=0.5$; dominance of one branch drives the order weight of the opposite branch toward unity without creating an additional free parameter.

Figure 9. Executed 3D endogenous dual-order surface. The vertical coordinate is $\omega_U^{\text{keep}}=\Lambda^\uparrow/(\Lambda^{\text{keep}}+\Lambda^\uparrow)$; $\omega_U^\uparrow$ is its exact complement. The surface is a mathematical visualization of reciprocal order generation, not a physical-energy measurement.

16.8 Dual-order non-expansion stress test

A deterministic two-mode normal form and a seeded random complex-matrix stress test were used to check the analytic bound in Equation (36). For the two-mode surface, $Y=\text{diag}(y_1, y_2)$, $D=(1, 1)/\sqrt{2}$, and the full space is used as the persistent subspace. The displayed grid spans $y_1, y_2 \in [10^{-2}, 10^2]$. Across that grid the maximum η_{DO} was 0.5765954, below the universal analytic bound $1/\sqrt{2} \approx 0.7071068$.

$$Y=\text{diag}(y_1, y_2), \quad D=(1, 1)^T/\sqrt{2}, \quad P_{\mathcal{W}^\wedge}(G)=I \quad (90)$$

$$\begin{aligned} \eta_{\text{DO}}(y_1, y_2) &:= \|\tilde{D}^\wedge\{\text{DO}\}\|_{\{G \oplus G\}} / \|D\|_G \\ &= [2\Lambda^{\text{keep}}\Lambda^\uparrow / ((\Lambda^{\text{keep}} + \Lambda^\uparrow)\Lambda_0)]^{1/2}, \quad \Lambda_0 := \|D\|_G^2 \end{aligned} \quad (91)$$

Figure 10. Executed 3D two-mode EDO coupling-gain surface η_{DO} . The displayed maximum is 0.5765954; the exact theorem gives $\eta_{\text{DO}} \leq 1/\sqrt{2}$ for every admitted state before the next closure rebuild.

The seeded stress test used 10,000 complex instances with dimensions $d=2, \dots, 8$, random positive loads $Y=A^\dagger A/d$, random complex states, and random persistent subspaces projected with $P_{\mathcal{W}^\wedge}(G)$. The maximum observed η_{DO} was 0.6412000821; no instance violated $1/\sqrt{2}$, and the largest numerical error in $\omega_U^{\text{keep}} + \omega_U^\uparrow = 1$ was 2.22×10^{-16} . These computations are sanity checks of the stated finite-dimensional construction. They do not prove natural-system realization and they do not identify Λ with physical energy.

16.9 Reciprocal-balance canonicity test

To test whether the reciprocal rule merely looks convenient, a one-parameter power family was compared with the exact balance condition. Let the keep weight be proportional to $(\Lambda^\uparrow)^\alpha$ and the upward weight to $(\Lambda^{\text{keep}})^\alpha$. The normalized balance residual is zero for all positive load ratios only at $\alpha=1$, which is precisely the reciprocal rule derived analytically from Equation (32). This numerical surface is illustrative; uniqueness itself follows algebraically from the two balance equations.

$$\omega_{\text{keep}}^\wedge(\alpha) := (\Lambda^\uparrow)^\alpha / [(\Lambda^{\text{keep}})^\alpha + (\Lambda^\uparrow)^\alpha], \quad (92)$$

$$\omega_\uparrow^\wedge(\alpha) := (\Lambda^{\text{keep}})^\alpha / [(\Lambda^{\text{keep}})^\alpha + (\Lambda^\uparrow)^\alpha]$$

$$\mathcal{R}_{\text{bal}}(\alpha) := |\omega_{\text{keep}}^\wedge(\alpha)\Lambda^{\text{keep}} - \omega_\uparrow^\wedge(\alpha)\Lambda^\uparrow| / [\omega_{\text{keep}}^\wedge(\alpha)\Lambda^{\text{keep}} + \omega_\uparrow^\wedge(\alpha)\Lambda^\uparrow] \quad (93)$$

Figure 11. Reciprocal-balance canonicity surface. Across eight decades of the load ratio, the normalized balance residual vanishes identically along $\alpha=1$. Other power exponents fail away from equal loads. The plot illustrates, rather than proves, the algebraic uniqueness of the reciprocal solution.

16.10 Canonical polar-transport stress test

The polar transport rule was tested on 10,000 seeded complex instances with dimensions 2–8. Each instance used a random positive master load, a random G -orthogonal persistent subspace, the exact neutral pair, reciprocal order weights, and the cross-fed analysis operator. The polar partial isometry satisfied $\|J^\dagger J - P_{\text{suppl}}\|_2 \leq 3.405 \times 10^{-15}$. Compression of an independently generated positive capacity remained positive in every case; the smallest observed

compressed eigenvalue was 0.00615466 . The largest Hermiticity defect was 1.019×10^{-15} . Reciprocal balance residuals stayed below 5.551×10^{-17} , the simplex-sum error below 2.220×10^{-16} , and the largest observed dual-coupler gain was $0.604756719 < 1/\sqrt{2}$. These are numerical checks of exact identities, not empirical evidence for a natural carrier.

Figure 12. Projective endogenous-order surface. The canonical keep weight depends only on the ratio $A\uparrow/A_{\text{keep}}$, demonstrating scale invariance and the one-degree-of-freedom character of the dual pair. Equal loads give $\omega_{\text{keep}} = \omega\uparrow = 1/2$ and projective scalar $r_U = 1$.

17. Interpretation of the simulations

The simulations support internal consequences of the Endogenous Dual-Order Strong Universal architecture only. They verify the exact neutral partition numerically, display a target-independent spectral formation boundary, reproduce the declared carrier order-feedback corridor, test shared-neutral integration and differentiation, check the resolvent derivative identity, exhibit bounded near-critical recursion, and directly stress-test the new reciprocal dual-order coupler.

The simulations do not prove natural universality. Their role is to establish calculational coherence and expose failure surfaces for the universal operator itself. The stronger empirical test is whether independent natural-system measurements map into the same invariant quotient and whether observed transitions commute with $\mathbb{U}$ within prospectively fixed error bounds.

18. Falsification architecture

A universal mathematical theory becomes vacuous if a failed realization can always be repaired by modifying the generator, externally adjusting order, or redefining the target class after inspection. The Endogenous Dual-Order Strong Universal Form therefore separates failures of the mathematical self-map, failures of carrier realization, and failures of empirical interpretation.

Table 4. Primary falsification surfaces of the EDO-URE program.

Failure surface	Decision condition
Formation failure	$\lambda_0 \geq 0$ or the Level-0 negative mode is not isolated.
Positive-geometry failure	A declared Gram-type construction or positivity domain fails, or $Y_{\{n+1\}}$ cannot be reconstructed as positive without changing the universal rule.
Neutral-resolution failure	Empirical response requires an independent pole or nonlinear law not derivable from the universal C_N/B_N family.
Persistence failure	The bounded non-decaying witness subspace is not closed/isolated on the declared carrier, or the required peripheral gap disappears under preregistered tolerances.
Generativity failure	A higher nonterminal iterate requires a target-specific capacity operator, gate, threshold, or synthesis map rather than the canonical polar transport and common formation rule.
Carrier-transfer failure	The empirical realization map must alter the universal generator or invariant quotient after entering a new domain.
Hierarchy-only null	A capacity-matched generic hierarchy predicts the same held-out transitions without BFG mediation/return structure.

Failure surface	Decision condition
Atomic failure	The proposed atomic readout adds no predictive or organizing content beyond relabeling established carrier theory.
Abiogenesis failure	Repeated-cycle closure does not depend on the predicted coupling, export, or witness relations under ablation.
Organism failure	Projection/readout/mediation variables add no prospective value beyond matched developmental nulls.
Consciousness failure	Matched recurrent or connectivity models reproduce held-out perturbational predictions without shared-neutral constraints.
Qualia failure	No non-circular content-specific closure invariant is found, or local carrier activity fully explains content and ownership.
Global no-retuning failure	Any target domain can be fit only by modifying the universal generator, invariant coordinates, quotient metric, or tolerance after seeing the outcome.
Endogenous-order failure	Observed or simulated generator behavior requires violating normalized reciprocal balance, adding an external level-specific order input, or introducing a branch-dependent coefficient not derivable from Equations (32)–(34).
Projection-gain failure	The canonical graph-metric projector or polar partial isometry cannot be defined on the declared active support, or the analytic non-expansion result fails under the stated assumptions.

19. Preregistration and no-retuning protocol

Every empirical realization test must freeze, before target outcomes are inspected, the carrier-to-BFG role extraction, neutral response family, bounded-nondecay witness criterion, peripheral-gap tolerance, positive graph-metric projection, reciprocal-balance readout, canonical polar transport, common formation gate, quotient metric, realization-map class, and all numerical tolerances. None of these objects may be replaced after the target domain or held-out outcome is known.

- Freeze operator family and numerical tolerances before outcome access.
- Separate structural level n from physical/developmental time t .
- Report retention, complement, export, spectral margin, and witness continuity jointly.
- Include hierarchy-only, order-independent, return-ablated, shuffled-coupling, and capacity-matched recurrent controls where applicable.
- Preserve negative outcomes and localize the failed bridge instead of rewriting the stage definition.
- For consciousness and qualia, preregister report-independent or report-minimized carrier estimators and construct the invariant class without using the phenomenal label as a feature.
- For abiogenesis, require repeated-cycle persistence rather than a one-off successful compartment.
- For cross-domain claims, use the same universal quotient coordinates and generator; only the empirical carrier map Π_s may differ, and its form must be declared before target outcomes are inspected. A scalar carrier order coordinate may be used only as a downstream readout of r_U or ω_U , never as a retuned input to $\mathbb{U}$.

20. Relationship to established science

The Endogenous Dual-Order Strong Universal Form is a mathematical closure architecture and a natural-realization research program, not a replacement for mature carrier theories. Matrix analysis and perturbation theory constrain the positive and spectral claims [7,8]; fixed-point theory remains relevant to carrier-level order consistency [11]; established physics constrains physical realization; chemistry and origin-of-life research constrain biological realization; and contemporary consciousness science constrains cognitive and phenomenal realization. The EDO-URE adds an internal order loop but does not remove those independent empirical burdens.

The burden is therefore sharper than explanatory compatibility. BFG must demonstrate that one prospectively frozen invariant quotient and one self-mapping operator, including the same reciprocal dual-order rule, compress, predict, or discriminate held-out natural data across domains better than capacity-matched alternatives without generator retuning.

21. Strong Universal closure certificate and scope separation

Within the declared BFG operator category, the principal mathematical sources of hidden freedom are now closed explicitly rather than listed as unresolved obligations. Positive geometry is generated as $K^\dagger K$ and rebuilt by a Gram construction; persistence is the bounded non-decaying recursive subspace; the Riesz contour is representational rather than selectable; dual order is the unique solution of normalized reciprocal balance; the next-level synthesis operator is the polar partial isometry of the cross-fed analysis map; terminal cases are absorbed by $\perp$; and the neutral core extends to separable Hilbert-space quadratic-form objects. What remains outside this closure certificate is not missing mathematics inside $\mathbb{U}$ but the empirical question of whether a given natural carrier realizes the declared category and quotient without retuning.

Table 5. Strong Universal closure certificate and empirical scope separation.

Item	Canonical Closure status
Dual-order origin	Closed mathematically: ω_U is the unique normalized solution of reciprocal load balance, not a fitted two-parameter law.
Persistence selection	Closed mathematically: $\mathcal{W}$ is defined by bounded non-decay; an isolated peripheral Riesz projector represents that subspace and is contour-independent.
Positive master geometry	Closed mathematically: $Y=K^\dagger K \geq 0$ and the next Y is regenerated by the same positive Gram construction.
Capacity transport	Closed mathematically on the active support: J is the polar partial isometry of the cross-fed analysis operator; no free next-level transport matrix is introduced.
Terminal cases	Closed mathematically: loss of dual support, persistence gap, positivity, or the common formation gate maps to the absorbing state $\perp$.
Infinite-dimensional carrier spaces	Closed at the resolvent level for separable Hilbert-space quadratic-form objects with closed densely defined K ; C_N and B_N remain bounded positive contractions.
Stage classification	Closed structurally: named natural stages do not occur in $\mathbb{U}$; the quotient is generated before any natural label is applied.
Dimensional units	Not part of the generator. Unit-bearing physical quantities are downstream carrier readouts and cannot alter $\mathbb{U}$.

Item	Canonical Closure status
Natural representation	Empirical realization hypothesis: Π_s must be fixed from measured carrier observables before transition outcomes are tested.
Atomic / chemical / biological realization	Empirical realization hypotheses: success requires fixed-rule commutation with $\mathbb{U}$ and failure cannot be repaired by generator retuning.
Consciousness / qualia realization	Empirical realization hypotheses: invariant-class membership does not by itself establish phenomenality; independent neural, behavioral, and report-linked tests are required.
Physical-energy identification	Explicitly not claimed by the structural closure loads Λ . A joule-valued energy readout requires an independently validated carrier map and conservation accounting.

22. Universal realization and validation program

The empirical program begins only after the mathematical closure certificate has been frozen. First publish the universal state schema, canonical persistence rule, reciprocal-balance rule, polar transport rule, positive rebuild, quotient metric, and numerical tolerances. Second generate invariant classes without natural labels. Third define carrier maps Π_s from independently measured observables. Fourth test transition commutation with $\mathbb{U}$. Fifth, where a scalar carrier order variable is used, test whether it is prospectively predicted by a fixed readout of r_U rather than fitted as an independent generator parameter.

This organization removes a common ambiguity in universal theories: empirical non-realization cannot be repaired by changing the mathematics after the fact, while mathematical closure cannot be presented as empirical confirmation. The universal theorem is therefore closed over its declared category, and the natural-universality claim is a collection of explicit falsifiable realization hypotheses. Scientific vulnerability is retained intentionally at the empirical boundary rather than hidden as an adjustable internal degree of freedom.

23. Canonical Endogenous Dual-Order Strong Universal Self-Closure Theorem

Conditional	Theorem	—	Canonical	Endogenous	Dual-Order	Strong	Universal	Self-Closure
Assume: (i) admissible Level-0 formation with an isolated determinate branch; (ii) exact neutral elimination with a nonnegative self-adjoint master load; (iii) the canonical bounded non-decay witness definition and an isolated peripheral spectral block; (iv) projection orthogonal in the positive graph metric; (v) reciprocal balance $\omega_{\text{keep}}\Lambda_{\text{keep}}=\omega\uparrow\Lambda\uparrow$ together with simplex normalization; (vi) polar support transport $Q=(\mathcal{A}DO\uparrow\mathcal{A}DO)^{\wedge\{1/2\}}$, $J=\mathcal{A}DOQ^{\wedge+}$; (vii) positive Gram-type reconstruction of the next master load; and (viii) one common formation gate with failure mapped to $\perp$. Then $\mathbb{U}:\tilde{X}_{\text{adm}}\rightarrow\tilde{X}_{\text{adm}}$ is a totalized self-map, the dual-order state is unique whenever both persistent loads are positive, the dual coupler obeys $\ \tilde{D}D\ ^2\leq\ D\ ^2/2$ in the master graph norm, the active-support transport is canonical up to the unique polar partial isometry, and positive master geometry is regenerated at every nonterminal iterate. The same neutral resolvent core remains valid for separable Hilbert-space quadratic-form objects generated by closed densely defined K . No natural-stage predicate, external order scalar, tunable persistence contour, or free next-level synthesis map is part of the generator. Natural realization and physical-energy identification are downstream empirical hypotheses and are not premises of the theorem.								

Proof sketch. Let X belong to the totalized admissible state category. If $X=\perp$, the absorbing definition gives $\mathbb{U}(X)=\perp$. Otherwise $Y\geq 0$ gives the positive graph metric and exact neutral pair. The witness space is defined by bounded non-decay, not by a target-selected contour; under the declared peripheral gap its Riesz representation is unique, and

Equation (29) gives the unique G-orthogonal projector. The exact neutral pair and metric projection yield nonnegative closure loads $\Lambda^{\{\text{keep}\}}, \Lambda^{\{\uparrow\}}$. Equation (32), together with simplex normalization, has the unique solution in Equation (33), proving that the dual-order pair is derived rather than freely chosen. Because C_N and B_N are commuting positive contractions and $C_N + B_N = I$, the cross-fed packet obeys Equation (36). The analysis operator in Equation (37) has a canonical polar decomposition; Equation (38) fixes the partial isometry on its active support, so the capacity compression in Equation (39) introduces no independent synthesis map. The positive Gram reconstruction yields $Y_{\{n+1\}} \geq 0$, and the common spectral formation gate either admits a nonterminal next state or maps the branch to $\perp$. The same resolvent identities extend to closed densely defined K on separable Hilbert spaces by Equations (45)–(46). Hence $\mathbb{U}$ is a level-independent self-map of the declared totalized operator category with endogenous order, canonical transport, positive rebuild, and no stage-specific fundamental response law. Natural-system realization remains an independent empirical proposition rather than an unproved assumption inside the theorem.

24. Conclusion

The Canonical Closure Edition strengthens the EDO Strong Universal Form by converting its most obvious discretionary choices into derived objects. The exact neutral pair generates two complementary persistent branches. Their graph-metric closure loads do not merely motivate a dual order rule; normalized reciprocal balance uniquely determines it. The resulting pair is a bounded projective representation of one load ratio, with balanced scalar coordinate $r_U = 1$, rather than two independent parameters. The persistent subspace is defined dynamically by bounded non-decay, and the next-level transport is fixed by the polar decomposition of the already determined cross-fed analysis operator.

The universal generator therefore contains neither named natural stages nor hidden carrier-specific order, contour, synthesis, or positivity choices. Positive geometry is rebuilt, failure is totalized, and the resolvent core extends beyond finite matrices through nonnegative self-adjoint quadratic forms. Physical, chemical, biological, cognitive, conscious, and phenomenal labels remain downstream hypotheses on target-independent invariant classes. This separation is intentional: it closes the mathematical architecture without converting mathematical closure into an empirical claim.

The resulting universality statement is correspondingly sharper. Within the declared admissible BFG operator category, one canonical map $\mathbb{U}$ regenerates the data required for its own next application. Across nature, the same statement becomes a prospective realization test: independently constructed carrier maps must commute with $\mathbb{U}$ without generator retuning. Failure of that test counts against natural universality; it does not license a new sector law. This is the strongest scientifically controlled form of the BFG universal-reclosure claim developed here.

Appendix A. Core formula register

Table 6. Compact formula register.

Role	Formula / statement
Level-0 source	$\mathcal{T}_0 = (\mathcal{D}_0, \mathcal{C}_0, \mathcal{N}_0); \mathcal{K}_0 = \mathcal{C}_0 + \mathcal{N}_0 - \mathcal{D}_0$
Formation	$\delta^*_{\pm} = \pm \sqrt{(-\lambda_0) \mathcal{C}_0}$ for simple $\lambda_0 < 0$
Positive load	$Y = A \dagger A \geq 0; Y_{\mathcal{C}} = L_{\mathcal{C}} \dagger H_{\mathcal{C}L_{\mathcal{C}}}$
Neutral retention	$C_{\mathcal{N}}(Y) = (I + Y)^{-1}$
Complement	$B_{\mathcal{N}}(Y) = Y(I + Y)^{-1} = I - C_{\mathcal{N}}(Y)$
Exact partition	$C_{\mathcal{N}} + B_{\mathcal{N}} = I$; quadratic retained+complement budget exact
Persistence	$R_{\mathcal{C}e_{\mathcal{T}}} = e_{\mathcal{T}}$; stable complement strictly contractive in a Lyapunov norm
Carrier order readout	$\varphi_{\text{obsv}}(\varphi_{\text{obs}}) = \Phi_{\text{src}}$ is a downstream carrier consistency relation; φ_{obs} must be mapped prospectively from $r_{\mathcal{U}}$ or $\omega_{\mathcal{U}}$ without entering $\mathcal{U}$.
Persistent metric subspace	$\mathcal{W}$ is the bounded non-decaying recursive subspace; isolated peripheral spectrum gives its unique Riesz representation. $G = I + Y$ defines the orthogonal projector $P_{\mathcal{W}^{\wedge}(G)}$.
Keep / upward / export	$P_{\mathcal{W}^{\wedge}(G)} C_{\mathcal{N}D}; P_{\mathcal{W}^{\wedge}(G)} B_{\mathcal{N}D}; (I - P_{\mathcal{W}^{\wedge}(G)}) B_{\mathcal{N}D}$.
Endogenous dual order	Reciprocal balance: $\omega_{\text{keep}} \Lambda_{\text{keep}} = \omega \uparrow \Lambda \uparrow$ and $\omega_{\text{keep}} + \omega \uparrow = 1$, hence $\omega_{\mathcal{U}} = (\Lambda \uparrow / S, \Lambda_{\text{keep}} / S)$; $r_{\mathcal{U}} = \Lambda \uparrow / \Lambda_{\text{keep}}$.
Cross-fed packet and bound	$\tilde{D}D O = \sqrt{\omega_{\text{keep}}} D_{\text{keep}} \oplus \sqrt{\omega \uparrow} D_{\text{up}}; \ \tilde{D}D O\ _{\mathcal{G} \oplus \mathcal{G}}^2 = 2 \Lambda_{\text{keep}} \Lambda \uparrow / S \leq \ D\ _{\mathcal{G}}^2 / 2$.
EDO-URE	$\mathcal{A}D O$ is fixed by neutral/persistent branches; $J = \mathcal{A}D O (\mathcal{A}D O \dagger \mathcal{A}D O)^{\wedge \{ \dagger 1/2 \}}$; $X_{\{n+1\}} = \mathbb{U}[X_n]$ after canonical capacity compression, positive rebuild, and the common gate.
Iteration	$\mathbb{U}: \bar{X}_{\text{adm}} \rightarrow \bar{X}_{\text{adm}}, \quad \mathbb{U}(\perp) = \perp, \quad X_n = \mathbb{U}^n[X_0],$ $\varphi_{\{U, n+1\}} = \mathbb{U}_{\mathcal{U}}[\square^{\wedge n}(X_0)].$
Cognitive load	$Y_{\varphi} = M_{\varphi} \dagger M_{\varphi}$
Cognitive integration	$\Xi_{\varphi} = \Sigma_{\{i \neq j\}} \ P_{iC_{\varphi}P_j}\ _{\mathcal{H}S}^2 / \ C_{\varphi}\ _{\mathcal{H}S}^2$
Differentiation	$\Delta_{\varphi} = \sigma_{\min}(C_{\varphi} H_{\text{act}})$
Universal invariant quotient	$Q_{\mathcal{U}} = \bar{X}_{\text{adm}} / \sim_{\mathcal{U}}$; natural labels enter only through external realization maps Π_s .

Appendix B. Simulation reproducibility register

Table 7. Exact numerical specification for executed figures.

Item	Specification
Software	Python 3.13.5; NumPy 2.3.5; Matplotlib 3.10.8.
Neutral surface	y_1, y_2 logarithmically sampled from 10^{-2} to 10^2 ; 160×160 grid; $\Gamma_{rc} = \frac{1}{2}(4c_1b_1 + 4c_2b_2)$.
Order-feedback surface	$\varphi \in [0.45, 1.55]$, $B \in [0.55, 0.98]$; $\alpha=0.35$, $\beta=0.25$, $d=0.45$, $c=0.08$; 180×180 grid.
Spectral gate normal form	$\delta \in [-1.6, 1.6]$, $g \in [0, 1.6]$; $\lambda_{\min} = 1 - \sqrt{\delta^2 + g^2}$; 180×180 grid.
Cognitive W	Unit-weight ring edges (1,2),(2,3),(3,4),(4,5),(5,1); chords $W_{13}=0.35$, $W_{24}=0.25$, $W_{35}=0.30$; normalized to $ W _2=1$.
Cognitive D	$\text{diag}(0.55, 0.65, 0.75, 0.85, 0.95)$.
Order fixed point	$\Phi_{\text{src}}=1$; $a=0.45$; $\varphi^*=0.7481333923$.
Integration surface	$\varphi \in [0.45, 1.25]$, $g \in [0, 1.8]$; 140×140 grid.
Recursive primitive	$R_0 = 1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$.
Near-critical gain	$r=0.995/\rho(R_0C_\varphi)=1.1422297734$ at $\varphi^*=0.7481333923$, $g=0.8$.
Derivative check	Centered finite difference $h=10^{-6}$; relative Frobenius error 4.2967×10^{-10} .
Trajectory	Initial $s_0=(0.90, -0.55, 0.35, 0.25, -0.20)$; 160 iterations; PCA obtained by SVD of centered 161-state matrix.
Dual-order surface	$\Lambda_{\text{keep}}, \Lambda^\uparrow$ sampled logarithmically from 10^{-2} to 10^2 on a 180×180 grid; $\omega_U^{\text{keep}} = \Lambda^\uparrow / (\Lambda_{\text{keep}} + \Lambda^\uparrow)$, $\omega_U^\uparrow = 1 - \omega_U^{\text{keep}}$.
Two-mode EDO gain surface	$Y = \text{diag}(y_1, y_2)$, $y_1, y_2 \in [10^{-2}, 10^2]$ logarithmic 180×180 grid; $D = (1, 1)/\sqrt{2}$; $P_{\mathcal{W}^\wedge(G)} = I$; $\eta_{\text{DO}} = \ \tilde{D}^{\wedge}\{\text{DO}\}\ _{\{G \oplus G\}} / \ D\ _G$.
Random complex EDO stress test	Seed 20260908; 10,000 instances; dimensions 2–8; $Y = A^\dagger A/d$ from complex Gaussian A ; random complex D and random subspace rank $1 \dots d$; G -orthogonal projection. Observed maximum $\eta_{\text{DO}} = 0.6412000821$; 0 violations of the analytic $1/\sqrt{2}$ bound; max simplex-sum error 2.22×10^{-16} .
Reciprocal-balance canonicity surface	$\log_{10}(\Lambda^\uparrow/\Lambda_{\text{keep}}) \in [-4, 4]$, $\alpha \in [0.2, 2.0]$; normalized residual $\mathcal{R}_{\text{bal}}$. The analytic zero-residual condition is $\alpha=1$.
Canonical polar-transport stress test	Seed 20260908; 10,000 complex instances; dimensions 2–8; random positive Y , random G -orthogonal subspaces, SVD polar factor. Max $\ J^\dagger J - P_{\text{suppl}}\ _2 = 3.405 \times 10^{-15}$; max balance residual $= 5.551 \times 10^{-17}$; max $\eta_{\text{DO}} = 0.604756719$.

Appendix C. Source-credit register

Table 8. Source-line provenance used by the Unification Paper.

Source line	Use in present paper
Marcel Theodor Wendé & Thomas W. Loker [3]	Co-authored Level-0 admissibility and pre-polar formation source line.
Marcel Theodor Wendé & Luke Kenneth Casson Leighton [4]	Recursive conjugate-closure source line $R_C=MRM^\dagger$, $M=\sqrt{N}$ and related stability geometry.
Canonical BFG source [1]	Universal positive closure architecture, exact neutral elimination, spectral gate, and LPCP.
Prior BFG sources [5,6]	Local order, fixed-point mediation, generativity/return architecture, and cross-scale compatibility.
Prior carrier-specific BFG sources [12,18,26,27,35,44]	Carrier-level structural constraints for atomic, biological, organismic, cognitive, phenomenal, and thermodynamic readouts.
This paper	Strong Universal Reclosure Equation, totalized admissible category, target-independent invariant quotient, positive-metric persistent projection, endogenous reciprocal dual-order construction, local dual-order non-expansion result, and the EDO-URE.

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Declarations

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Data availability. No external empirical dataset was analyzed. Numerical results are synthetic deterministic or seeded calculations from equations and parameters fully stated in the manuscript.

Code availability. The simulation equations, parameter ranges, random seed, software versions, and deterministic initial conditions required to reproduce the displayed figures are provided in Appendix B. A public repository release may be attached to a later archival version.

Ethics. No human participants, animal experiments, or identifiable personal data were used.

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